

Pathway Enrichment on the Recommended Model's Survival-Active Gene Programs

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1. Motivation and direction convention

The recommended model (YFB, DeSurv-aligned gene selection, $K=7$, no cohort indicator; mean external C-index = 0.627 across 5 held-out PDAC cohorts) uses only 2 of its 7 latent gene expression programs for survival prediction ($K_{\text{eff}} = 2$), a result confirmed by three independent methods (warm-start, deflation-init, joint (K, α) Bayesian optimization; see DECISIONS.md 2026-07-13). This report assigns biology to those two programs — **Program 7** and **Program 3** — via gene-set enrichment and four independent lines of orthogonal evidence.

Direction convention (DECISIONS.md, 2026-06-16). Adverse/protective direction is read from the *marginal* survival association of each program's projection score, not the joint model's β sign — the two disagree here (joint $\hat{\beta}_7 = -0.041$, $\hat{\beta}_3 = +0.011$; a suppression effect among correlated programs). By the marginal convention: **Program 7 = Adverse**, **Program 3 = Protective**. All labels and directional predictions below use this convention.

2. Methods

Gene weights come from the D4 fit's $\mathbf{F}$ matrix (2064 genes $\times$ 7 programs; a point-exponential prior makes every weight ≥ 0). Primary enrichment method: fgsea (scoreType="pos", one-sided by construction since weights are non-negative), ranking all 2064 genes by weight per program. Confirmatory method: over-representation analysis (clusterProfiler::enricher) on the top-N weighted genes ($N \in \{50, 100, 150\}$), background = the same 2064-gene universe (not the whole genome, since these genes are already survival/variance-selected). Gene-set collections: MSigDB Hallmark, Reactome (CP:REACTOME), KEGG (CP:KEGG_MEDICUS), GO Biological Process (GO:BP) via msigdb 26.1.0 (collection/subcollection arguments; category/subcategory are deprecated as of this version), plus a custom PDAC collection: Moffitt basal/classical (25+25 genes; Moffitt et al. 2015), Bailey 4-subtype (Squamous/Immunogenic/Pancreatic Progenitor/ADEX; 87/145/73/57 genes; Bailey et al. 2016), and DeSurv's own D1/D2/D3 factor gene lists (270 genes each, extracted from the DeSurv SI appendix, Tables S7-S9; Young et al., PNAS 2026). All 7 programs were enriched (a sanity check that the 5 non-survival-active programs enrich for generic/distinct-but-non-prognostic biology, not nothing); reporting below focuses on Programs 3 and 7. set.seed(1) before all fgsea calls; BH-adjusted p (padj) per collection, reported at padj < 0.10.

3. Gene-set enrichment (F1, T1, C1)

Table 1: T1: all sets enriched at padj<0.10 for Programs 3 and 7 (KEGG_MEDICUS_REFERENCE_/REACTOME_ prefixes stripped for width; full names in F1 and the underlying CSV).

program	label	collection	set	size	NES	padj
3	Protective	PDAC_custom	DeSurv D1 ClassicalTumor	249	1.06	0.01700
3	Protective	PDAC_custom	DeSurv D3 BasalLikeTumor	259	1.06	0.01700
3	Protective	KEGG	ITGA B FAK CDC42 SIGNALING PATHWAY	14	1.28	0.02300
3	Protective	KEGG	ITGA B FAK RAC SIGNALING PATHWAY	14	1.28	0.02300
3	Protective	KEGG	ITGA B RHOGAP RHOA SIGNALING PATHWAY	14	1.28	0.02300
3	Protective	PDAC_custom	Moffitt Classical	21	1.17	0.03500
7	Adverse	PDAC_custom	DeSurv D3 BasalLikeTumor	259	1.08	0.00061
7	Adverse	KEGG	GF RTK PI3K SIGNALING PATHWAY	33	1.23	0.00110
7	Adverse	KEGG	GF RTK RAS PI3K SIGNALING PATHWAY	33	1.23	0.00110
7	Adverse	KEGG	GF RTK RAS ERK SIGNALING PATHWAY	31	1.23	0.00120
7	Adverse	Reactome	DEVELOPMENTAL CELL LINEAGES	98	1.13	0.01000
7	Adverse	PDAC_custom	Bailey Squamous	28	1.16	0.03000
7	Adverse	Reactome	NEGATIVE REGULATION OF THE PI3K AKT NETWORK	30	1.22	0.07800

Program 7's headline associations are basal-like/squamous tumor identity (DeSurv D3, Bailey Squamous), MET/EGFR-driven RTK signaling (three related KEGG pathways sharing MET, EGFR, AREG, EREG, TGFA, VEGFA), and a basal keratin/integrin developmental program (Reactome). Program 3's headline association is classical/differentiated tumor identity (DeSurv D1, Moffitt Classical: GATA6, HNF4A, TFF1-3, AGR2/3) — with a secondary, weaker (padj=0.017 vs 0.035; see §6) hit on the DeSurv basal-like list, discussed honestly rather than omitted.

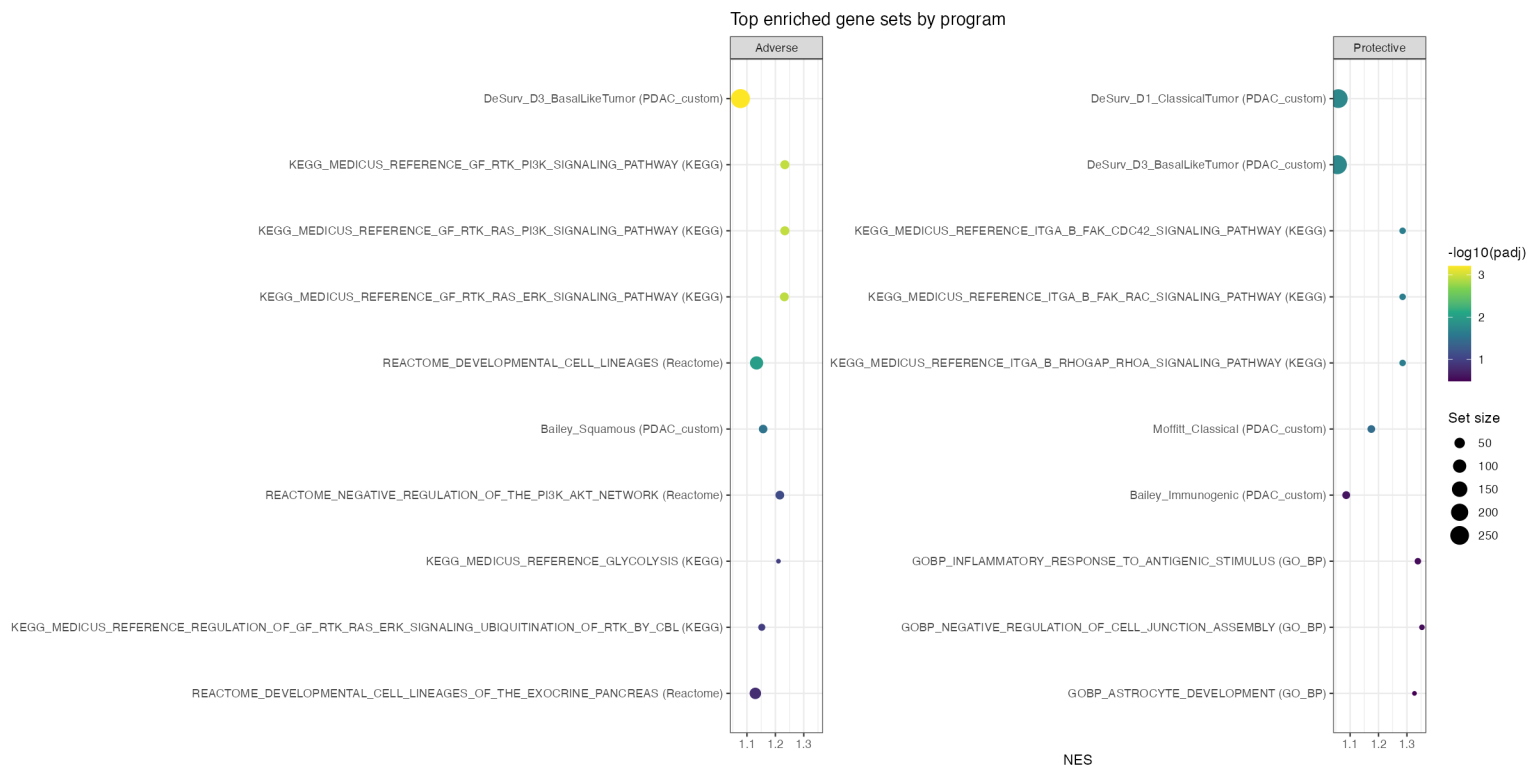


Figure 1: Top enriched gene sets per active program. Size = gene-set size; color = $-\log_{10}(\text{padj})$.

Table 2: C1: coherent-hit count (fgsea sets at $\text{padj} < 0.10$) per program, all 7. The 3 fully inert programs (1, 2, 4) show 0 hits; Programs 5-6 (nominally survival-inactive) still enrich for real stromal/immune biology – genuine variance axes in the data that are simply not survival-predictive in this model, not noise.

program	label	n_coherent_hits
1	Inactive	0
2	Inactive	0
3	Protective	6
4	Inactive	0
5	Inactive	12
6	Inactive	40
7	Adverse	7

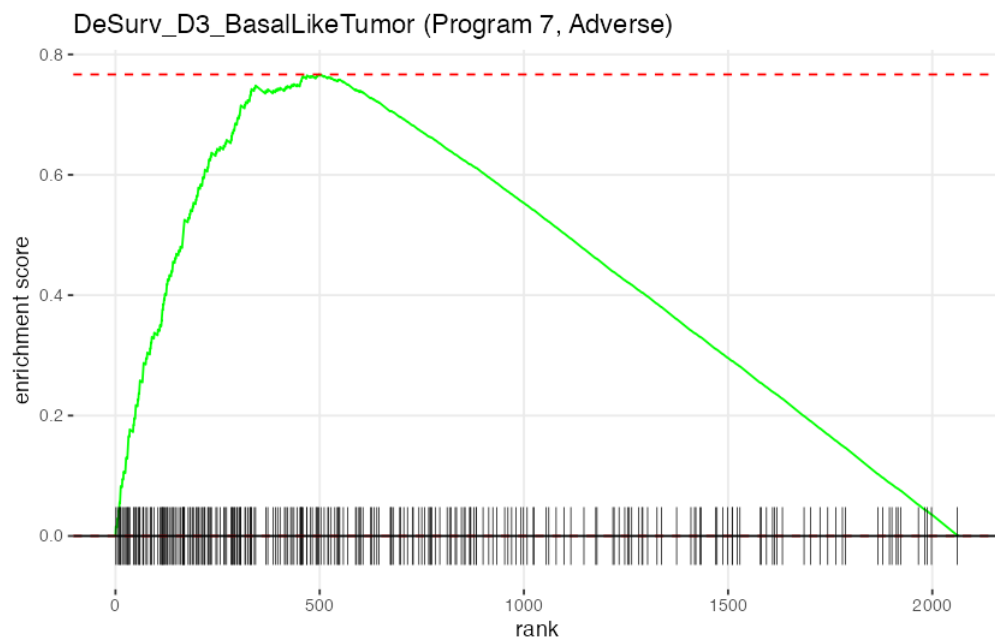


Figure 2: Running enrichment score, Program 7 vs. DeSurv D3 (Basal-like Tumor).

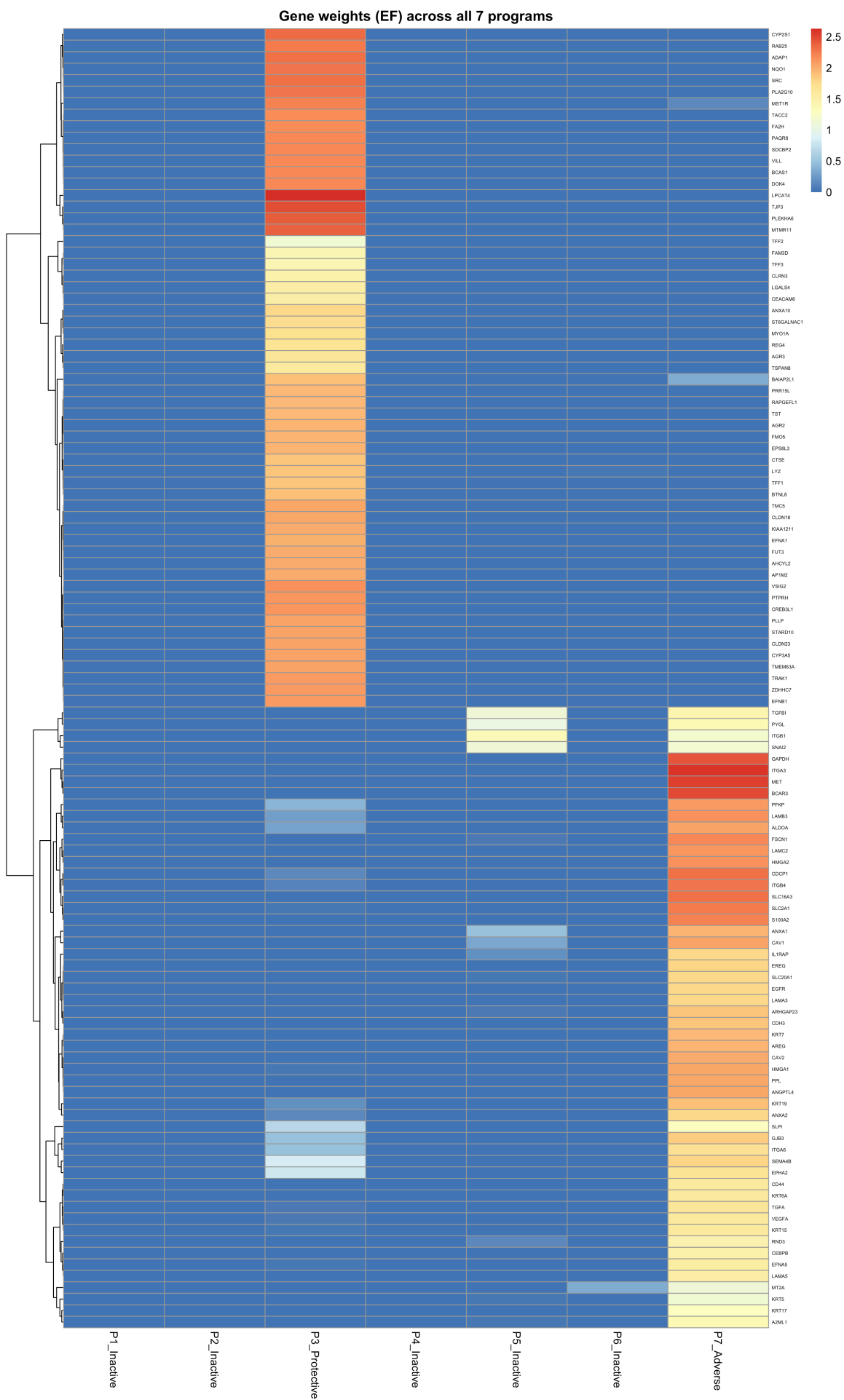


Figure 3: Gene weights across all 7 programs for the union of headline leading-edge genes. Clean block-diagonal structure: Program 3's and Program 7's genes are nonzero almost exclusively in their own column.

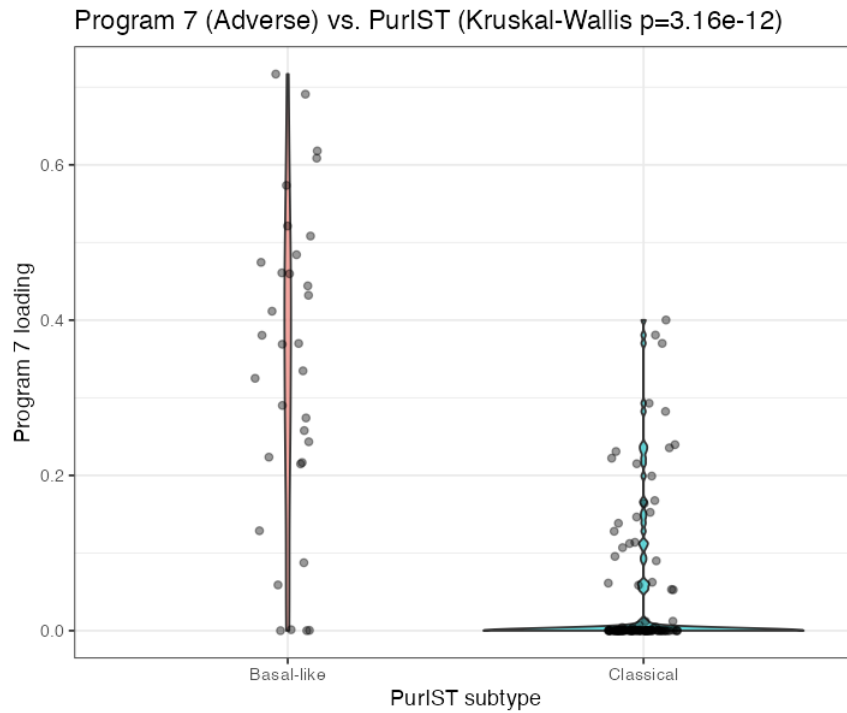


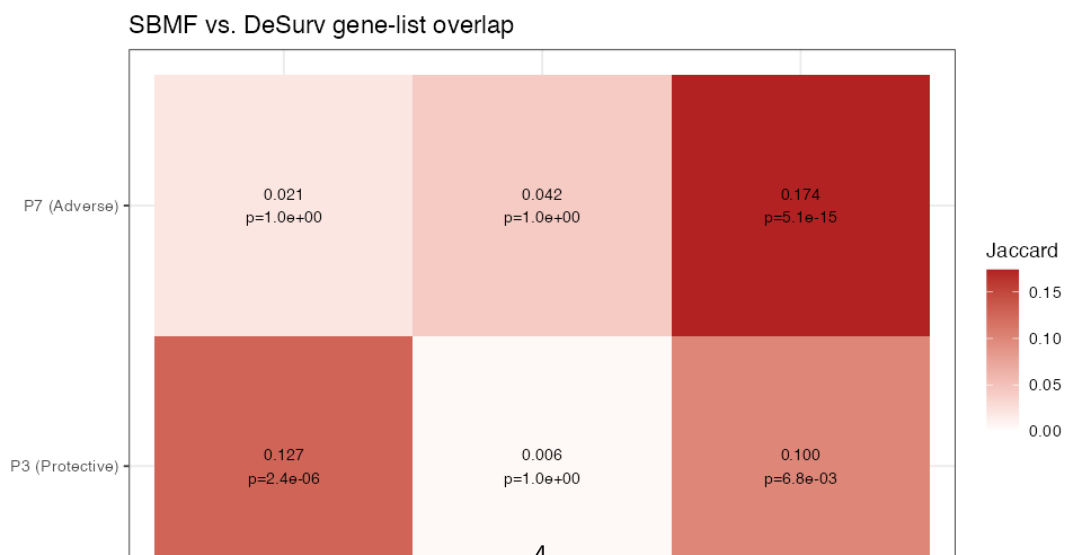
Figure 4: Program 7 loading by PurlST subtype (Program 3's companion figure is in the output directory).

Table 4: C2: per-cohort hazard ratio and C-index for each program's headline leading-edge signature.

program	label	cohort	n	HR	p	cindex
3	Protective	Dijk	90	0.84	0.3300	0.426
3	Protective	Moffitt_GEO_array	123	0.88	0.5500	0.492
3	Protective	PACA_AU_array	63	0.49	0.0360	0.387
3	Protective	PACA_AU_seq	52	0.42	0.0120	0.392
3	Protective	Puleo_array	288	0.68	0.0074	0.422
7	Adverse	Dijk	90	1.99	0.0046	0.597
7	Adverse	Moffitt_GEO_array	123	1.30	0.1900	0.543
7	Adverse	PACA_AU_array	63	1.70	0.2200	0.594
7	Adverse	PACA_AU_seq	52	2.40	0.0730	0.587
7	Adverse	Puleo_array	288	1.56	0.0048	0.603

Program 7's signature has $HR > 1$ (worse survival) in **5/5** external cohorts (range 1.30-2.40, individually significant in 2/5); Program 3's has $HR < 1$ (better survival) in **5/5** cohorts (range 0.42-0.88, individually significant in 3/5). This exceeds the plan's own bar of "a majority of cohorts" for both programs.

6. SBMF vs. DeSurv gene-list overlap (F5, T4)



7. Direction sanity check (C3) and conclusion

Four independent methods were run: gene-set enrichment (§3), PurlST subtype concordance (§4), external-cohort survival (§5), and direct SBF-DeSurv gene overlap (§6). **All four agree, with no contradictions:** Program 7 (Adverse) is a basal-like/squamous, MET/EGFR-RTK-driven, EMT-associated program — the PDAC subtype literature's established worse-prognosis axis. Program 3 (Protective) is a classical, differentiated-epithelial program — the established better-prognosis axis. The one honest nuance (Program 3's secondary, weaker basal-like association, plausibly driven by shared general epithelial-identity markers rather than basal-specific drivers) is reproduced by two independent methods (§3, §6) and is reported rather than smoothed over, but does not contradict the dominant classical/protective signal.

Reproducibility. fgsea 1.36.2 (permutation seed = 1); msigdb 26.1.0; clusterProfiler 4.18.4; org.Hs.eg.db 3.22.0. Per-gene-set source citations and MSigDB-universe mapping rates recorded in `results/benchmark_sim/outputs/pathway_enrichment/genesets_manifest.txt`. All producing scripts: `run_pathway_enrichment.R` (Steps 5-6), `run_subtype_concordance.R` (Step 7), `run_external_cohort_robustness.R` (Step 8), `run_sbf_desurv_overlap.R` (Step 9), all under `results/benchmark_sim/`.